RKDF University

Ph.D Entrance Examination Syllabus

Electrical Engineering

MEASURING INSTRUMENTS

Principle of operation and construction of PMMC, MI, Dynamometer, Induction, Thermal and Rectifier type instruments – Measurement of voltage and current – use of ammeter shunts and voltmeter multiplier – Us e of CT and PT for extending instrument ranges.

POWR SYSTEM

Various system of transmission & their comparison, HVDC transmission Converter, inverter, filters & substation layout. Voltage and Reactive Power control. Types of supports, types of conductors, types of insulators, their properties, selection and testing, , , HVDC transmission Converter, inverter, filters & substation layout. Voltage and Reactive Power control. Distribution Systems: Primary and secondary distribution systems, concentrated & uniformly distributed loads on distributors fed at one and both ends, ring distribution, sub mains and tapered mains, voltage drop and power loss calculations, voltage regulators, Feeders Kelvin’s law

ELECTRICAL MACHINE

Constructional features of DC machine , Principle o f operation of DC generator,EMF equation ,Types of excitation, No load and load characteristics of DC generators ,commutation , armature reaction. Principle of operation of DC motors, Back emf , Torq ue equation ,Types of DC motors-Speed ,Torque characteristics of DC motors ,Starting of DC motor s: 2 point starter, 3 point starter, 4 point starter , Speed control , Losses and efficiency ,Applications Principle of operation , Constructional features of single phase and three phase transformers , EMF equation ,Transformer on No load and Load ,Phasor diagram ,equivalent circuit ,Regulation

SWITCHGEAR AND PROTECTION

Introduction to protective relaying,classification of relays , over current relays , directional over current relays , differential relays,distance relays , frequency relays-negative sequence relays -

Introduction to static relays , comparison of electromagnetic and static relays, Bucholz Relay. Theory of arcing and arc quenching-RRRV-current chopping-capacitive current breaking-DC circuit breaking switchgear, fault clearing and interruption of current-Breakers.

CONTRO SYSTEM

Transient and steady state response , definitions , mathematical expression for standard test signals , type and order of systems , step response of first order and second order under damped systems. Time domain specifications of second order under damped systems , Step response of second order critically damped and over damped systems.

ELECTRIC DRIVES

Individual and collective drives- electrical braking, plugging, rheostatic and regenerative braking load equalization use of fly wheel criteria for selection of motors for various industrial drives, calculation of electrical loads for refrigeration and air-conditioning, intermittent loading and temperature rise curve. Starting methods of three phase induction motor , C ogging & Crawling ,Speed control , Voltage control , Rotor resistance control , Pole changing , Frequenc y control , Slip – energy recovery scheme , Double cage rotor , Induction generator ,Synchronous induction motor.

POWER ELECTRONICS

Basic structure & switching characteristics of power diodes, Power transistor & SCR, Triggering methods of SCR, R, RC, and UJT firing circuits for SCR, series and parallel operation of SCR, need for snubber circuits, di/dt & dv/dt protection. Introduction to Triac, GTO, MOSFET, IGBT, FCT and MCT

RENEWABLE ENERGY SYSTEMS

Energy Sources, Comparison of Conventional and non-conventional, renewable and non-renewable sources. Statistics of world resources and data on different sources globally and in Indian context. Significance of renewable sources and their exploitation. Energy planning, Energy efficiency and management.